

ISRM Whitepaper

Why an operational reference model for intelligent systems is needed

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Executive summary

Modern AI standardization is rich in terminology, management systems, lifecycle processes, risk management, quality and domain-specific evaluation. At the same time, AI-enabled agents and cyber-physical systems increasingly perform closed-loop action: they receive goals or events, construct context, generate options, make authority-constrained decisions, execute effects and observe outcomes. ISRM proposes a common operational vocabulary for that runtime chain.

The proposal is intentionally broader than machine learning and narrower than a universal systems architecture. Its unit of analysis is a Processing Episode within a declared system boundary. The model is designed to make responsibility, uncertainty, authority, execution status and recursive delegation explicit.

1 Problem statement

- The same word “agent” can describe a chatbot, tool-using LLM, robot, workflow orchestrator or deterministic controller.
- Current implementations often blur recommendation, decision and execution, creating audit and safety ambiguity.
- Multi-agent delegation can obscure where authority originated and whether it expanded.
- AI outputs can express confidence poorly; downstream components may silently treat uncertain context as fact.
- Operational effects are frequently logged as success when only an API attempt is known.
- Learning/adaptation is often mixed with ordinary state/context updates.

2 Proposed contribution

ISRM contributes a technology-neutral functional decomposition plus testable invariants. The intended standardization value is not the novelty of each individual concept, but interoperability of architectural reasoning: two organizations should be able to map different implementations into the same responsibility model and identify where authority, uncertainty, provenance and effects cross boundaries.

3 Why six layers

Layer	Question answered
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Intent	What outcome is sought?
Perception	What was observed?
Context	What state is relevant now?
Cognition	What could be done or inferred?
Decision	What is permitted/selected now?
Execution	What operation was attempted?

Environment is deliberately excluded because it is the object of observation/effect, not an internal responsibility. Control, Trust and Adaptation are planes because each constrains multiple layers rather than representing a single sequential transformation.

4 Differentiation from prior art

Prior work	Core idea	ISRM distinction / relationship
OODA	Observe–Orient–Decide–Act loop for decision/action under uncertainty.	ISRM decomposes orientation into Context/Cognition, separates authorization from execution, adds explicit system boundary, trust/control/adaptation planes, recursive delegation and conformance objects.
BDI	Belief–Desire–Intention model of rational agency.	ISRM is not a mental-state model; it focuses on runtime responsibility, authority, execution evidence and environment feedback across AI and non-AI implementations.
NIST 4D/RCS	Hierarchical sensory processing, world modeling, value judgment and behavior generation with feedback.	Strong architectural predecessor. ISRM is domain-neutral, non-hierarchical by default, explicitly separates Decision/Execution, adds authority/trust/adaptation planes, recursive systems and conformance classes.
MAPE-K	Monitor–Analyze–Plan–Execute over shared Knowledge for autonomic systems.	ISRM adds explicit Intent, Decision authorization, Perception/Context distinction, authority/provenance invariants, multi-system recursion and effect confirmation semantics.
ISO/IEC 23053	Framework for ML-based AI systems.	ISRM is implementation-neutral and focuses on runtime intent/event-to-effect responsibilities.
ISO/IEC 5338	AI lifecycle processes.	ISRM is runtime architecture, not lifecycle process.
ISO/IEC 42001 / 23894 / NIST AI RMF	Management and risk frameworks.	ISRM supplies architectural attachment points for controls; it does not replace risk

		management.
ISO/IEC/IEEE 42010	Architecture-description requirements.	Complementary: 42010 describes how architectures are documented; ISRM proposes a specific reference model to document.
IEEE P3225.05	Reference architecture for Internet of Agents.	Potential overlap; ISRM is broader than agent networking and includes deterministic/cyber-physical systems. Coordination with this work is essential.

5 Falsifiability

A useful reference model must be rejectable. ISRM should be revised or abandoned if independent mappings repeatedly require ad-hoc layers, if Decision/Execution cannot be distinguished in consequential systems, if the same responsibility maps inconsistently across domains, or if the proposed invariants cannot be objectively tested.

6 Expected benefits

- Common architecture vocabulary across AI, robotics and automation.
- Clearer human-in-the-loop and delegated-authority design.
- Better incident reconstruction: proposal vs decision vs execution vs confirmed effect.
- Technology-neutral conformance profiles.
- A bridge between runtime architecture and existing governance/risk/safety standards.

7 Standardization hypothesis

The proposed standards need is a reference model and vocabulary for operational intelligent-system responsibility and traceability. A DIN project could initially be national or could be developed as German input to ISO/IEC JTC 1/SC 42. The competent DIN body is expected to determine the final route and scope.

8 Authorship and standardization

This whitepaper records Marcel Reitz as originator of the ISRM proposal. If a DIN/ISO/IEC/IEEE project is accepted, the resulting consensus document will be produced under the rules of the standards body and may differ materially. Origin of the proposal and authorship/ownership of a later standard are distinct.

References

DIN Normungsantrag: <https://www.din.de/de/mitwirken/normungsantrag>

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